FIT3080 Applied Class on Propositional and First-order Logic

Exercise 1: Propositional logic – models, validity and satisfiability

Consider a vocabulary with only four propositions A, B, C and D. How many models are there in the following sentences? Are these sentences valid, satisfiable (but not valid) or unsatisfiable?

(a) $B \vee C$

(b) $\neg A \vee \neg B \vee \neg C \vee \neg D$

Exercise 2: Propositional logic – validity and satisfiability

Consider the following sentence:

$[(Food \Rightarrow Party) \vee (Drinks \Rightarrow Party)] \Rightarrow [(Food \wedge Drink) \Rightarrow Party]$

Convert this sentence to CNF, and determine whether it is valid, satisfiable (but not valid) or unsatisfiable.

Exercise 3: Validity and satisfiability

Prove that $\alpha \models \beta$ iff $\alpha \wedge \neg\beta$ is unsatisfiable. This is the main theorem for resolution-refutation.

Exercise 4: Propositional Logic – Horn clauses, forward and backward reasoning

Consider the following statements:

1. If Susan gets high marks she is successful.
2. If Susan is bright and studies she will get high marks.
3. If Susan is not bright she won't pass the subject.
4. If Susan is diligent she will study.
5. If Susan is not diligent she will have fun.
6. If Susan has fun she will not study.
7. Susan is diligent.
8. Susan has passed the subject.

(a) Using the propositions H (high marks), SL (successful), B (bright), S (study), P (pass), D (diligent) and F (have fun), translate the above statements to Propositional Logic. Which of these clauses can be converted to Horn clauses?

(b) Apply backward reasoning to prove that Susan is successful.

(c) Apply forward reasoning to prove that Susan is successful.

Exercise 5: First-order Logic – Equivalence

Prove or argue that the following equivalences hold (Slide 48):

(a) $\forall x[P(x) \wedge Q(x)] \equiv \forall x P(x) \wedge \forall y Q(y)$

(b) $\exists x[P(x) \vee Q(x)] \equiv \exists x P(x) \vee \exists y Q(y)$

Exercise 6: First-order Logic – Unification

In the following pairs of expressions, w, x, y and z are variables, A and B are constants, and f and g are functions. Do these pairs of expressions unify? If they do, give the substitution that unifies them. If they don't, then indicate where the unification fails.

(a) $P(x, f(x))$ and $P(f(A), f(A))$

(b) $P(x, f(x), f(x))$ and $P(y, z, f(y))$

Exercise 7: First-order Logic – Resolution refutation

Consider the following statements:

1. John likes all food.
2. Anything that one eats and isn't killed by is food.
3. Bill eats peanuts.
4. Bill is still alive.

(a) Represent these statements as predicate calculus formulas using the predicates: LIKES, FOOD, EATS and ALIVE.

(b) Convert these statements to clauses.

(c) Use resolution-refutation to prove that John likes peanuts.